

Simulation studies on multi-armed bandit with imputation

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I. INTRODUCTION

In medical community, rapid progress in machine learning improves clinical decision nowadays. Bandit algorithm is one of the reinforcement learning and the goal of bandit is to maximize the cumulative reward for each round t . Let A_t be an action at time t and R_t is the reward for selecting A_t . At first we try A_1 and get reward R_1 and next time, we use R_1 to make a decision for second round A_2 , and so on. ($A_1 \rightarrow R_1 \rightarrow A_2 \rightarrow R_2 \rightarrow \dots \rightarrow A_{1000} \rightarrow R_{1000}$).

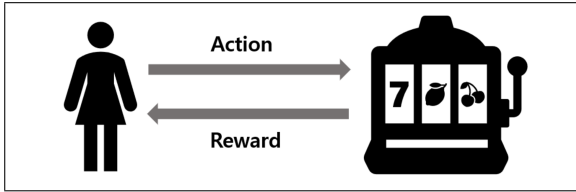


Fig. 1. Bandit algorithm

In biostatistics, 'arms' and 'rewards' correspond to 'clinical trial' and 'success or fail of that trial'. Multi-Armed Bandit Problem (MABP) can be used about what clinical trials be allocated to patients. The goal is to strike an appropriate balance between allocating the best treatment ever performed on a patient (exploitation) and the other treatment that has not been performed but is expected to show better clinical results (exploration).

The key problem when using MAB in clinical trials is that information should be updated using response information to determine which treatment to allocate to the next patient. However, it is impossible to update for various reasons (e.g. do not appear, adverse reaction, death). Then the subsequent allocations on patient could be affected due to the missing and these missing data reduces the statistical power of a study and produces biased parameter estimates. Therefore, the imputation for the missing part is necessary.

"Some performance considerations when using multi-armed bandit algorithms in the presence of missing data" [2] is the main literature that I used in project. They compared how missing data affects each of the various bandit algorithms and in-

troduced how the results of the bandit changed after using mean imputation method. In particularly, they focused on two-armed bandits with binary outcomes and applied different probabilities of missingness in both arms through simulation. After that, they evaluated the performance of different algorithms using proportion of patients assigned to the experimental arm (p^*).

My main goal of the project is as follows. First, do simulation studies on various multi-armed bandit problems (Current Belief, Thompson sampling, UCB, etc) with missing data since there is no official code in the paper. Next, apply mean imputation method to simulated bandit results introduced in this paper. And last, analyze the results of simulations and compare with original figure in this paper.

II. MAIN FRAMEWORK

2.1 Allocation procedure of bandit algorithms in the presence of missing data

Let consider a general two-armed bandit model with binary responses in the presence of missing data under missing at random that the probability of missing is the same within groups. Let arm 0 is a control and arm 1 is an experimental clinical trial. Patients will be sequentially assigned to these arms and p_k would be the true probability of success in arm k . We will test a null hypothesis $H_0 : p_0 = p_1$ against a one-sided alternative hypothesis $H_A : p_0 < p_1$. There exist missing responses with a probability $p_k^m \leq 0.5$ and total number of missing responses, successes and failures on arm k at time t are denoted by $M_{k,t}$, $S_{k,t}$ and $F_{k,t}$. With an uniform prior ($s_{k,0} = 1, f_{k,0} = 1$ which implies equal allocation in initial state, the observed data $x_{k,t} = (s_{k,0} + S_{k,t}, f_{k,0} + F_{k,t})$ has an impact on the next allocation. Note that $N_{k,t} = M_{k,t} + S_{k,t} + F_{k,t}$, our focus is $p^* = \frac{N_{1,n}}{n}$ which is the proportion of patients assigned to the experimental arm. The allocation procedure of bandit algorithms is introduced in 2. The allocation simulation results of my own for each bandit algorithm, which will be introduced in the next chapter 2.2, are attached to figure 3.

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if observed ( $p_{k,t}^m < u$ ,  $u \sim U_{[0,1]}$ ) then
   $M_{k,t} += 0$ ;
  if failure ( $p_k < u$ ,  $u \sim U_{[0,1]}$ ) then
     $F_{k,t} += 1$ ;
     $S_{k,t} += 0$ ;
  else
     $S_{k,t} += 1$ ;
     $F_{k,t} += 0$ ;
  end
else
   $M_{k,t} += 1$ ;  $F_{k,t} += 0$ ;  $S_{k,t} += 0$ ;
end
Update  $x_{k,t}$  with ( $S_{k,t}$ ,  $F_{k,t}$ ) for the next allocation

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Fig. 2. Algorithm of allocation procedure

2.2 Bandit algorithms

In the paper [2], the author introduces 9 different bandit algorithms and among them, I would use 4 bandit algorithms below and the whole algorithms are summarized in Table 1. The initial value of $\pi_{k,0}$ and $I_{k,t}$ is 0.5, which means the first assignment is an equal allocation for all algorithms. In more details, $c = \frac{t}{2n}$ in TTS and $c = 1$ in RTS. $\hat{\mu}_{k,t}^B = \frac{s_{k,0} + S_{k,t}}{s_{k,0} + S_{k,t} + f_{k,0} + F_{k,t}}$ and $\beta = \sqrt{2 \log(t)}$, $\lambda_k(t) = \frac{1}{\sqrt{s_{k,0} + S_{k,t} + f_{k,0} + F_{k,t}}}$.

- Randomized algorithms (TTS, RTS): The patient at time t is randomly assigned to arm k with probability $\pi_{k,t}$.
- Deterministic algorithms (CB, UCB): The patient at time t is deterministically assigned to the arm k with the larger index values $I_{k,t}$.

Algorithm	$\pi_{k,t}$ or $I_{k,t}$
TTS	$\pi_{k,t} = \frac{P(\max_i p_i = p_k)^c}{\sum_{k=1}^K P(\max_i p_i = p_k)^c}$
RTS	$\pi_{k,t} = \frac{P(\max_i p_i = p_k)^c}{\sum_{k=1}^K P(\max_i p_i = p_k)^c}$
CB	$I_{k,t} = \hat{\mu}_{k,t}^B$
UCB	$I_{k,t} = \hat{\mu}_{k,t}^B + \beta \cdot \lambda_k(t)$

Table 1. Bandit algorithms

- Tuned Thompson Sampling (TTS)

The probability $\pi_{k,t}$ is proportional to $P(\max_i p_i = p_k)$ i.e., posterior probability that treatment k is the best arm given data. Since the variable $S_{k,t}$ is distributed on Bernoulli distribution with probability p_k , the posterior distribution of p_k is beta distribution: ($p_k \sim \text{Beta}(s_{k,0} + S_{k,t}, f_{k,0} + F_{k,t})$). [3] [1] In early stage of allocation there is not enough data, and posterior probability is so variable that this method might take an interior treatment. The tuning parameter $\frac{t}{2n}$ can

help avoiding wrong selection. The figure 3 is the result of TTS implementation over single simulation under the null ($n = 200, p_0 = p_1 = 0.7$). It tends to select arm even when there is no treatment difference between two arms.

- Raw Thompson Sampling (RTS)

RTS is raw version of TS and the tuning parameter of TTS is fixed at $c = 1$ in RTS. Compared with TTS, selection under RTS happens earlier than TTS in the figure 3. The reason is that RTS is more deterministic with fixed tuning parameter than TTS, even though both are randomized algorithms.

- Current Belief (CB)

The reward $I_{k,t} = \hat{\mu}_{k,t}$, which is the mean posterior probability of success. In CB, once one arm is selected, all of the next enrolled patients will be assigned to this arm, and the index value $I_{k,t}$ of the unselected arm will not change anymore since this method only aims to exploitation. This algorithm might make a wrong selection and make many allocations to the inferior arm.

- Upper Confidence Bound index (UCB)

UCB explores the arm with higher uncertainty because of higher information gain represented by the term beta times $\lambda_k(t)$. $\lambda_k(t)$ is the confidence interval of standard deviation of $\hat{\mu}_{k,t}$. Since $\beta \cdot \lambda_k(t)$ goes from infinity to zero, the index values have an initial rapid increase for the first few patients but then decrease in both arms like figure 3.

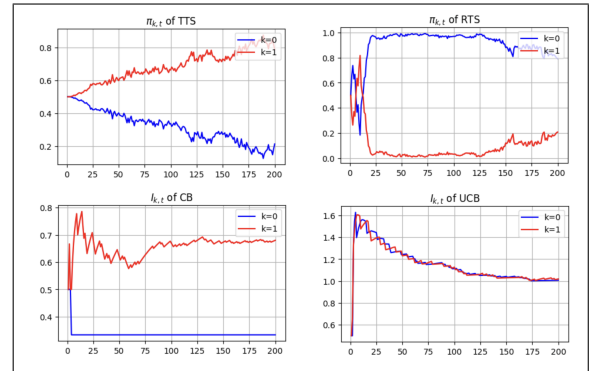


Fig. 3. Performance of allocation procedure

2.3 Mean imputation approach

Mean imputation is a single imputation that replaces missing values with the mean value of that variable. In this problem, $\hat{p}_{k,t} = \frac{s_{k,0} + S_{k,t}}{s_{k,0} + S_{k,t} + f_{k,0} + F_{k,t}}$ is as estimation of true success probability p_k . Then a random value is drawn from Bernoulli($\hat{p}_{k,t}$) to substitute missing outcome. The initial value is $\hat{p}_{k,0} = 0.5$, meaning that the treatment has an prior success probability of 50%.

III. RESULTS

To investigate the impact of different probabilities of missingness, I tried 6 scenarios 2, first 3 under the null ($H_0 : p_0 = p_1$) and last 3 under the alternative. These scenarios repeated 10,000 replications except for TTS and RTS, which uses 1,000 replications for computational reasons. And then I investigated several combinations of missing probabilities. Case (a) $p_0^m = p_1^m = p^m \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5\}$ is equal probabilities of missingness in each arm, case (b) $p_0^m \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5\}$, $p_1^m = 0$ is missingness only in the control, and case (c) $p_0^m = 0$, $p_1^m \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5\}$ is missingness only in the experimental arm.

Scenario	(p_0, p_1)	Trial size n
S1	(0.10, 0.10)	200
S3	(0.50, 0.50)	200
S5	(0.90, 0.90)	200
S7	(0.10, 0.30)	162
S9	(0.40, 0.60)	254
S11	(0.70, 0.90)	162

Table 2. Simulation studies

My own simulation results of $\mathbb{E}[p^*]$ based on different probabilities of missing under the null and alternative are attached in figure 4. Grey lines, blue lines, and red lines are correspond to case (a), (b) and (c), respectively. Solid lines correspond to the results without mean imputation, while the dashed lines correspond to the results with mean imputation.

- Tuned Thompson Sampling (TTS): Under the null, the impact due to the missing responses is relatively low as a result of exploration. Under the alternative, approximately 70% – 80% of patients are assigned to the experimental arm and the results of simulation are robust across different scenarios. Also, mean imputation solve missing problem due to properties of exploration
- Raw Thompson Sampling (RTS) : The shape of figure is similiar to TTS overall, but RTS is slightly affected by missing under extreme scenarios. Due to the fixed tuning parameter, it is slightly more exploitative than TTS. Under the null, the impact due to the missing responses is relatively low except for $p^0 = p^1 = 0.9$. Under the alternative, approximately 80% – 90% of patients are assigned to the experimental arm. The $\mathbb{E}[p^*]$ ratio is higher than TTS due to early selection.
- Current Belief (CB) : In almost every scenario, CB is largely affected by missing because of the

exploitative properties. Under the null, if p_k is small, CB allocates more patients to missing arm because it takes a long time to select arm. If p_k is large, CB allocates less patients to missing arm because of an early selection. Under the alternative, it performs differently across the scenarios because the deterministic properties depends on p_k . If p_k is large, CB performs worse than small p_k because of early selection. Also, due to this nature, mean imputation has no effect on large p_k .

- Upper Confidence Bound index (UCB) : The results are robust across all scenarios and mean imputation could solve the impact of missingness due to the explorative nature. UCB allocates more patients to the missing arm because $I_{k,t}$ decreases as the selection accumulates.

IV. CONCLUSION

To investigate the impact of missing and the effect of mean imputation on bandit algorithms for various scenarios, I simulated bandit mechanism introduced in the paper [2] and uploaded the Python code for whole procedure in github.com/jdy5294/bandit-with-missing.

The results obtained through simulation are summarized as follows. First, explorative algorithms tend to assign more in the missing arm, while exploitative algorithms are less likely to select the arm with missing data. In specifically, TTS and RTS (explorative and randomized) are not affected by missing. UCB (explorative and deterministic) allocates more patients to missing arm but the impact of missing could be mitigated by mean imputation. CB (exploitative) is largely affected by missing responses and could not be alleviated by a mean imputation approach.

Therefore, it can be concluded that the presence of missing responses can adversely affect the correct decision when the bandit algorithm is applied to the clinical trials. According to the simulation results, the result can be improved by using the imputation approach instead of ignoring the missing case. However, the input method assumes missing at random (MAR), there is a limitation that it is difficult to apply it to an actual missing situation where the actual MAR is not assumed. I also tried IPW or DR but it was not successful so I couldn't bring it to the result. If these methods success, I think we can get better result.

REFERENCES

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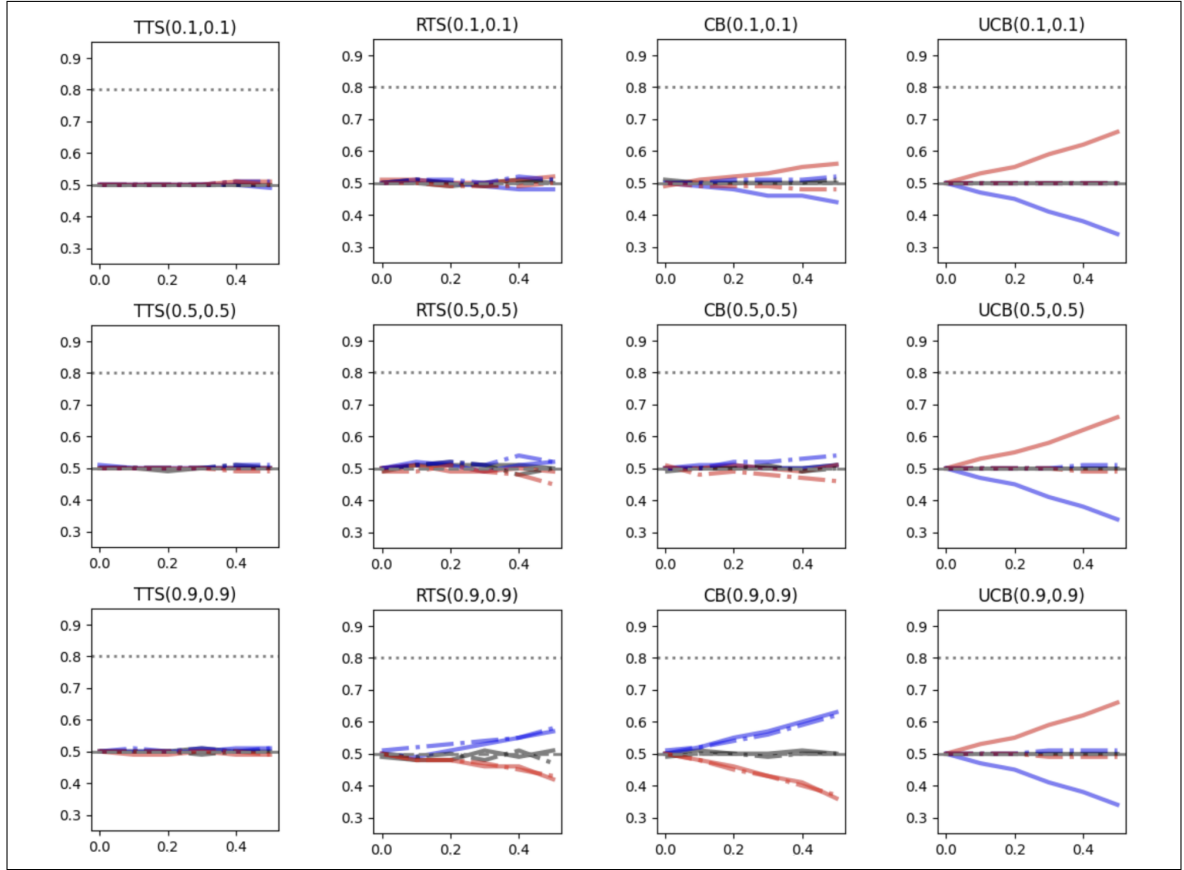


Fig. 4. Simulation under the null and alternative : "method name"(p^0, p^1)